

Ramratan Padhy

+91 7735334802 | ramratanpadhy59@gmail.com
github.com/consolelogram | linkedin.com/in/ramratan-ntlap

EDUCATION

SRM University AP

Bachelor of Technology in Computer Science Engineering

- **Year 1** | SGPA: 9.0/10.0

Amaravati, Andhra Pradesh

2025 – Present

EXPERIENCE

Next Tech Lab AP

Associate

Amaravati, Andhra Pradesh

Oct 2025 – Present

- Research associate at a QS ReImagine awarded student society.
- Collaborating with a research collective to explore low-level implementations of emerging technologies.
- Contributing to peer-to-peer technical workshops on systems programming and AI architecture.

Notecove (Stealth Startup)

Technical Content Writer

Remote

May 2025 – Aug 2025

- Authored 30+ comprehensive technical articles and guides on emerging software trends and AI ethics.
- Synthesized complex engineering concepts into accessible documentation for a specialized audience.
- Collaborated with the founding team to define the technical voice and content strategy for the platform.

Hush (Startup)

ML Intern

Remote

March 2026 – Present

- Worked with image models to discern user profiles to keep the app authenticity.
- Collaborated with the founding team to define and enhance user experience.

PROJECTS

Grand Tribunal | *Python, Modal, Nemotron, VoiceCPM*

Completed

- Shipped an Ace Attorney-style turn-based AI debate game utilizing voice input for an immersive, hands-free player experience. Trained and deployed two specialized Nemotron models on character personality datasets to serve as dynamic opponents and logical judges, interfaced through a custom Gradio frontend.

High-Performance Editorial Portfolio | *HTML, CSS, Web Optimization*

Completed

- Architected a zero-dependency, single-file portfolio site optimized for instant edge delivery, achieving sub-300ms load times and a perfect 100/100 Lighthouse performance score.
- Engineered a completely static, JavaScript-free architecture that inherently eliminates third-party vulnerabilities, maximizing security while ensuring optimal crawlability for modern AEO and SEO bots.

AXtheR (Windows) | *C, Windows API, Wintun, Cryptography*

Completed

- Developed a high-performance, zero-dependency, easy to run UDP VPN for Windows by integrating the Wintun kernel driver and Winsock2.
- Engineered a zero-heap allocation data path with multi-threaded asynchronous I/O and dynamic Windows routing table manipulation.
- Implemented a self-contained, RFC-compliant ChaCha20-Poly1305 cryptographic engine from scratch in C with constant-time MAC verification.

Independent Research on BitNet-1.58b | *Python, Machine Learning, Mathematics*

Completed

- Conducting novel research and experimentation on BitNet-1.58b by altering its context length.

Temporal RAG pipeline | *Python, RAG, ETL*

Completed

- Developed an end-to-end pipeline where you can input a video, and make any query regarding it.
- Worked with advanced data structures and various audio-video technologies to produce a system capable of answers queries about a video based on timeline and not only semantical similarity.

TECHNICAL SKILLS

Languages: Python, C, LaTeX, Godot (GDScript)

Developer Tools: APIs, Vector Databases, PyTorch, Scikit-Learn, TensorFlow, Windows API, AntigraVity

Core Concepts: Data Structures, Transformer Architectures, LLaMA Architecture, RAG, Technical Documentation, Mechanistic Interpretability, Model Evaluation

Soft Skills: Technical Communication, Research Synthesis, Problem Solving, Teamwork